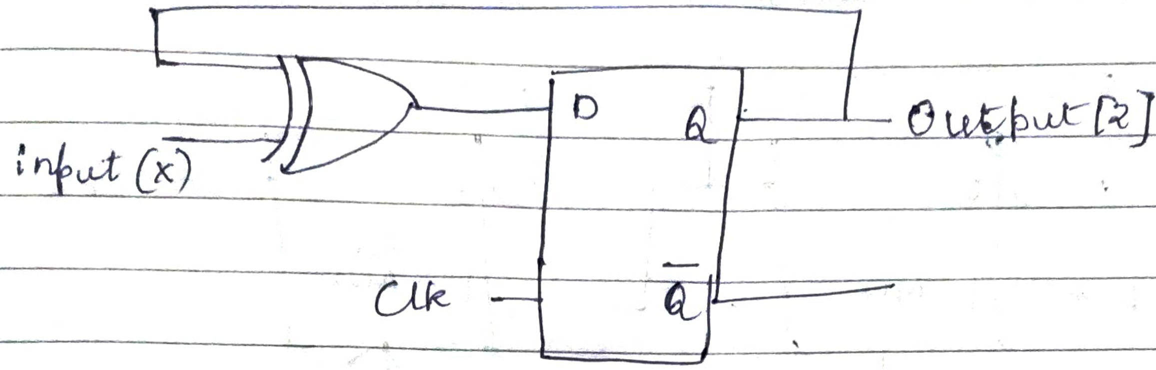
Assignment 4.

1. Explain the behaviour of given circuit:



XOR gate inputs are
external i/p = x.
feedback i/p = Q.

State table:

Present (Q)	input (x)	Qnext
0	0	0
0	1	1
1	0	1
1	1	0

$D = X \oplus Q$.

$Q_{next} = D$.

$Q_{next} = X \oplus Q$.

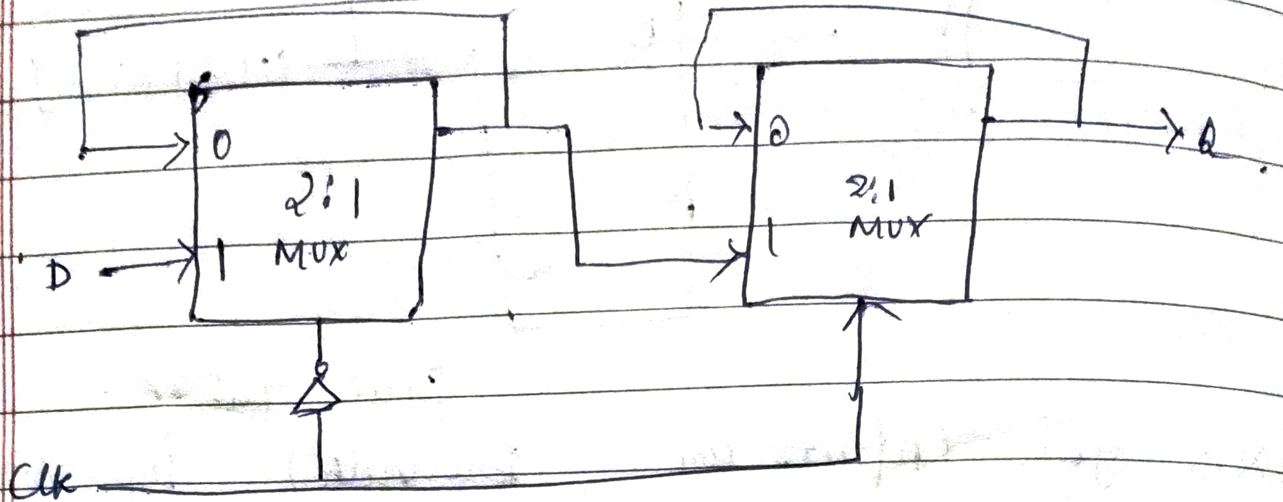
$Z_{next} = X \oplus Z$.

Case 1 : $x = 0$: $Q_{next} = Q$.

Case 2 : $x = 1$ $Q_{next} = \bar{Q}$

Circuit is TFF

2. Design a posedge triggered DFF using 2:1 Mux only.



$Q_{next} = Clk \cdot 0 + \overline{Clk} \cdot Q$.

Operation :

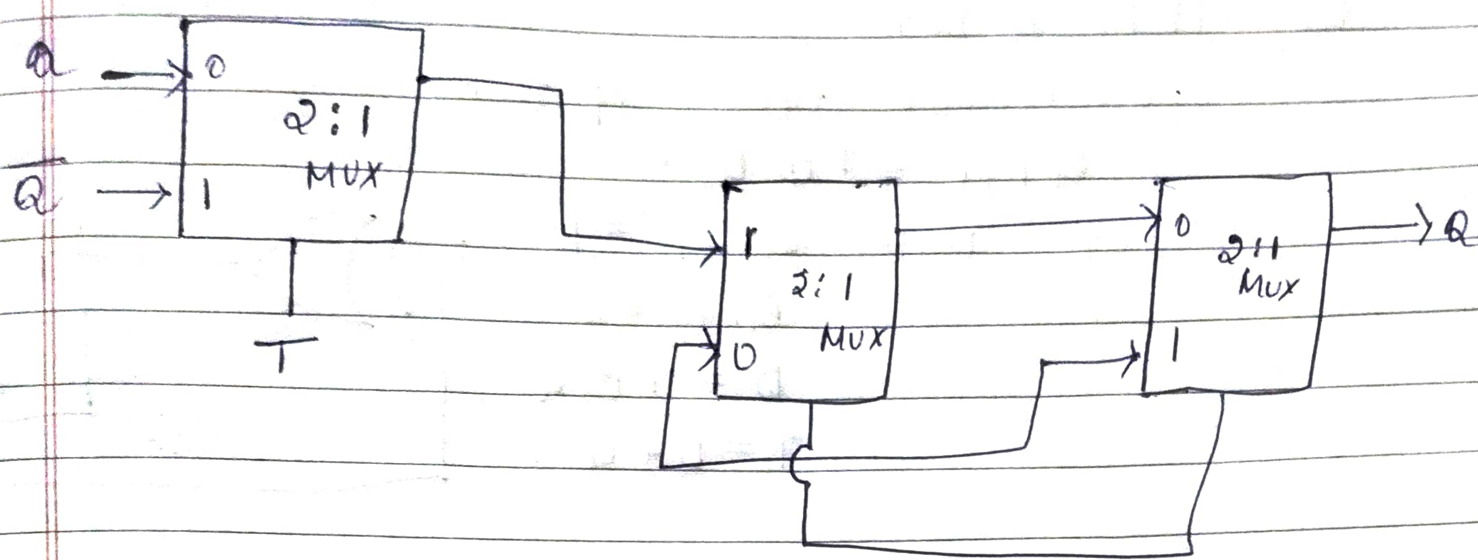
Master enable = $\overline{Clk}$

Slave enable = Clk.

Clk = 0 master tracks i/p, slave holds

Clk = 1, master slaves copies masters output.

3. Design a negedge FF triggered TFF using 2:1 MUX



4. Convert by adding external gates

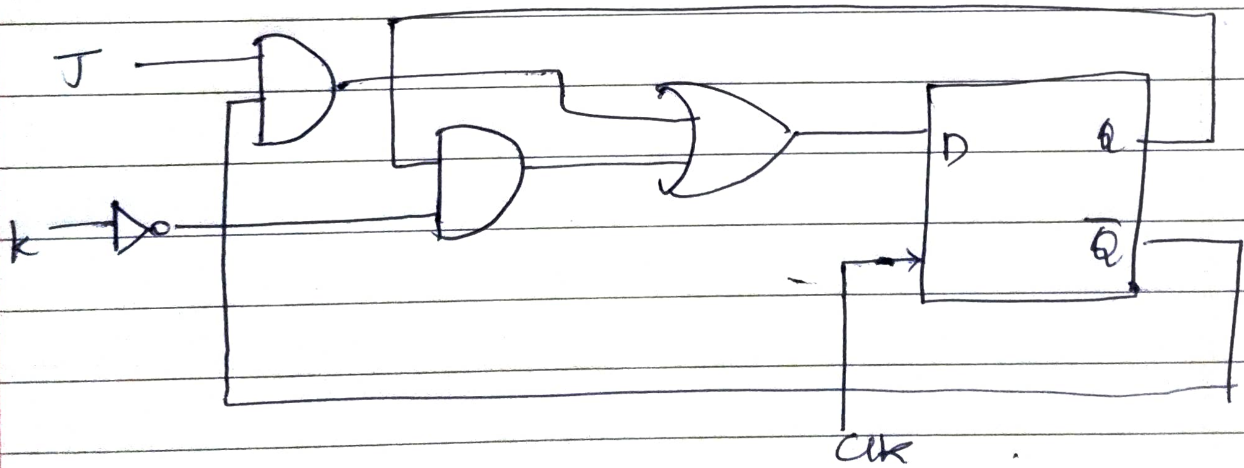
a) A DFF to JKFF

Characteristic equation of JK

$$Q_{n+1} = J\bar{Q} + \bar{K}Q.$$

For DFF $D = Q_{n+1}$

$$D = J\bar{Q} + \bar{K}Q.$$



b) TFF to DFF

Characteristic eqn of TFF.

$$Q_{n+1} = T \oplus Q .$$

For DFF $Q_{n+1} = D$

$$Q_{n+1} = D$$

$$D = T \oplus Q$$

$$T = D + Q .$$